

SRM University AP

B.Tech CSE – 6th Semester

Subject: Automata and Compilers Design

Course Code: CSE 304

CLA-3

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Section:Z

Roll No:AP23110010905

SRM University AP
B.Tech CSE – 6th Semester
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Name: DANDA LASYA Section: Z Roll No: AP23110010905

Last Date for submission: 20/04/2026 Max Marks: 15

1. Consider the following grammar and give the syntax directed definitions to constructed parse tree. For the input expression $4*7+1*2$ construct an annotated parse tree according to your syntax directed definition. [CO-4] [5 marks]

$E \rightarrow E * T \mid T$

$T \rightarrow T + F \mid F$

$F \rightarrow \text{digit}$

2. Write the quadruple, triple, indirect triple for the statement $a := b * -c + b * -c$. [CO-4] [5 marks]

3. Prepare a short video presentation on any topic from the course syllabus. Record yourself explaining the chosen topic clearly.

Upload the video either to your Google Drive or to YouTube, and ensure that the access permissions are set appropriately (viewable by the instructor).

Submit the video link as part of your assignment. [CO-1 to CO-5] [5 marks]

QUESTION 1:-

Automata & Compilers Design

Course Code: CSE304

CLA-3.

Name: - Danda Lasya Section: - Z Roll No: AP23110010905

1) Consider the following grammar and give the syntax directed definitions to construct parse tree. For the input expression $4 * 7 + 1 * 2$ construct an annotated parse tree according to your syntax directed definition.

$$E \rightarrow E * T \mid T$$

$$T \rightarrow T + F \mid F$$

$$F \rightarrow \text{digit}$$

• Syntax Directed Definition

$$E \rightarrow E_1 * T \quad \{E.val = E_1.val * T.val\}$$

$$E \rightarrow T \quad \{E.val = T.val\}$$

$$T \rightarrow T_1 + F \quad \{T.val = T_1.val + F.val\}$$

$$T \rightarrow F \quad \{T.val = F.val\}$$

$$F \rightarrow \text{digit} \quad \{F.val = \text{digit.lexval}\}$$

Input expression: - $4 * 7 + 1 * 2$

Step by step evaluation:-

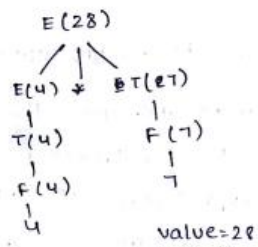
$$4 * 7 = 28$$

$$1 * 2 = 2$$

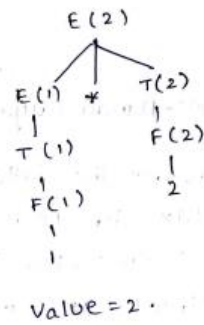
$$28 + 2 = 30$$

Valid partial parse tree-

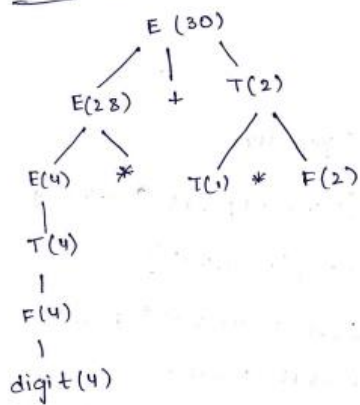
For $4 * 7$



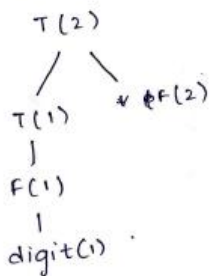
For $1 * 2$



Annotated Parse Tree



Right subtree:-



QUESTION 2:-

2. write a quadruple, triple, indirect triple for the statement $a := b * -c + b * -c$.

Step 1: Break into subexpressions

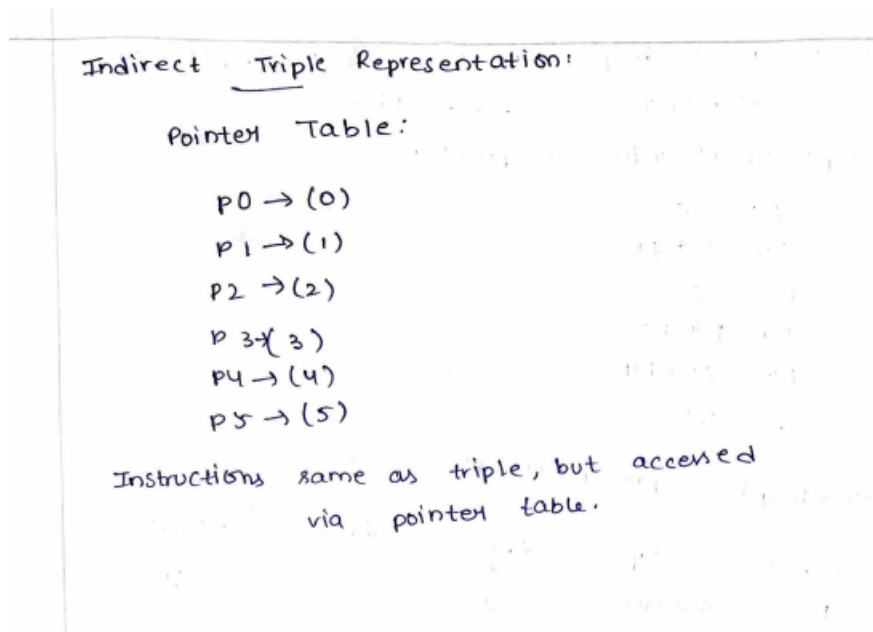
$t1 = -c$
 $t2 = b * t1$
 $t3 = -c$
 $t4 = b * t3$
 $t5 = t2 + t4$
 $a = t5$

Quadruple representation:-

Index	Op	Arg 1	Arg 2	Result
1	uminus	c	-	t1
2	*	b	t1	t2
3	uminus	c	-	t3
4	*	b	t3	t4
5	+	t2	t4	t5
6	:=	t5	-	a

Triple representation:-

Index	Op	Arg 1	Arg 2
0	uminus	c	-
1	*	b	(0)
2	uminus	c	-
3	*	b	(2)
4	+	(1)	(3)
5	:=	(4)	a



LINK FOR 3RD QUESTION:-

<https://drive.google.com/drive/folders/1oYz-RACTQecMB7PgJZEGdD4IhIQRQhOW?usp=sharing>

